

Hanzhang (Henry) YIN

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EDUCATION

Brown University | Providence, RI **Sep. 2025 – Jun. 2027**

- **Degree:** Sc.M. in Computer Science
- **GPA:** 4.0 / 4.0 | Department of Computer Science

University of Rochester | Rochester, NY **Aug. 2021 – Jun. 2025**

- **Dual Degree:** B.S. in Computer Science & B.S. in Applied Mathematics
- **GPA:** 3.77 / 4.0 | Hajim School of Engineering
- **Honors & Awards:** Dean's List

RESEARCH EXPERIENCE

Operational Logic for Interactive Articulated Objects | Advisor: *Prof. Daniel Ritchie* **Sep. 2025 – Present**

- Developed IOL, a portable asset-level representation for interactive articulated objects, encoding latches, buttons, pedals, timers, interlocks, joint coupling, and multi-step state-dependent operation.
- Implemented an object-local update mediator that intercepts simulator-proposed joint updates and permits, blocks, clamps, couples, or augments them using grounded propositions, temporal constraints, and conditioned effects.
- Built an LLM-assisted authoring pipeline with controller synthesis and semantic-trace repair, and evaluated IOL on operational object traces and embodied interaction tasks requiring valid object operation.

PRISM: PD End-of-Life Prognostication | Advisor: *Prof. Jiebo Luo* **Feb. 2024 – Feb. 2026**

- Developed PRISM, a Parkinson's-disease-specific machine learning model for 6-month end-of-life prognostication among nursing home residents, addressing the lack of empirically validated PD-specific hospice prediction tools.
- Built and evaluated traditional and advanced models, including Logistic Regression, LightGBM, Random Forest, Naïve Bayes, and Bi-LSTM, across Admission, LongStay, and Panel time-series datasets in collaboration with clinicians at URM.
- Compared model performance against Medicare hospice guideline baselines using AUC, sensitivity, specificity, precision, recall, and F1-score, and analyzed feature-selection patterns to identify clinically interpretable mortality-risk signals.

Image Compression and Generation Research | Advisor: *Prof. Yan Wang* **Jul. 2024 – Oct. 2024**

- Investigated combining DCVC with Diffusion Posterior Sampling (DPS) for video codecs.
- Evaluated performance on ImageNet-1k using FID and PSNR; explored a codec structure with enhanced denoising and generative capabilities for images/videos.

INTERNSHIP

KeeperAI | *Software Development Engineer Intern* **Jun. 2023 – Aug. 2023**

- Developed and enhanced front-end webpage designs using JavaScript and Fluent UI, integrating external NLP AI chatbot plugins to facilitate event scheduling within the Microsoft Teams app.
- Resolved security vulnerabilities in the user login verification module by redesigning regular expressions, improving the efficiency of account verification between the website and database APIs.

PROJECTS

Sand2016 – MPM Sand Simulation with Drucker–Prager Elastoplasticity **Apr. 2026**

- Co-developed a physics-based sand simulator by extending the MATTER MPM framework with a Klar et al.-style Drucker–Prager elastoplasticity model for granular material animation.
- Implemented sand plasticity components including return mapping, per-particle hardening state, friction-angle control, cone-parameter projection, and scene presets for hourglass flow, pouring piles, stick-drag trenches, notched-block collapse, chute, and funnel tests.

Amazing Terrain - Real-Time Procedural Environment Renderer **Dec. 2025**

- Implemented a real-time procedural environment renderer featuring fBm/ridged-noise terrain generation, multi-layer PBR material blending, normal mapping, tri-planar texture projection, and animated water.
- Implemented L-System vegetation with randomized turtle grammar, biome-aware tree placement, and OpenGL instanced

rendering for high-density branch/leaf geometry using shared meshes.

- Co-developed a cinematic atmosphere and post-processing stack with rain/snow particle billboards, exponential distance fog, HDR off-screen rendering, color grading presets, and Bézier-spline camera paths with quaternion interpolation.

Belief-Aware Multi-Agent RL Agent for Partially Observable Battle Arena

Dec. 2025

- Designed a belief-aware agent for a partially observable multi-agent grid-combat environment with latent strengths, simultaneous actions, stochastic combat outcomes, and global combat broadcasts.
- Combined Beta-Bernoulli strength posteriors, pairwise combat belief estimation, compact relational state abstraction, and bias-risk-shaped tabular Q-learning for robust policy learning under uncertainty.
- Evaluated against baseline and ablated agents across mixed-policy and custom-only arenas, analyzing the contribution of beliefs, heuristics, and Q-learning to survival and reward.

Mini-CLEVR VQA Experiments

Apr. 2025

- Built a lightweight VQA benchmark on the synthetic Mini-CLEVR dataset; released a data generator and a clean JSONL schema with reproducible training scripts.
- Implemented two compact baselines: ResNet-18 + SBERT and CLIP ViT-B/32 + LoRA (r=8); designed a simple fusion head ($\text{concat} \oplus \text{mix} \oplus \|\text{diff}\|$) for fast ablations.
- Ran ablation studies on feature unfreezing (e.g., layer4), LoRA rank, and fusion strategies; documented results with checkpoints and dataset links.

Simple QR Code Scanner & Decoder

Apr. 2024

- Implemented image preprocessing (e.g., Sauvola binarization) to enhance QR code detection; used size ratio and connected-component analysis to identify finder patterns.
- Established code bounding boxes by forming a right-angle triangle base from the three corner squares to determine location and orientation.
- Decoded QR contents (URLs, text, contacts) and achieved ~ 20 FPS real-time display; applied a Kalman filter to mitigate jitter for stable tracking.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, JavaScript/TypeScript, SQL, Rust, OCaml
- **Graphics / Simulation:** OpenGL, GLSL, Qt, Three.js, Mesh Processing, Path Tracing, ..
- **ML / AI:** PyTorch, scikit-learn, OpenCV, CNNs/RNNs/Transformers, reinforcement learning, ...
- **Systems / Tools:** VScode, Pycharm, Linux, Git, Vim, React, Flask, Socket.IO, W&B,...